

Section 1 Introduction to Statistics

Statistics I · Fall 2026 · Zong-Wei Yeh · Department of Finance, CCU

Overview

Almost every important decision is made without complete information. A fund manager chooses a portfolio before knowing next year's returns. A firm launches a product without asking every possible buyer. Both can still collect some data. But the data cover only part of the group they want to understand. That part is a sample. The decision must then be made from the sample.

Statistics is the subject that lets us defend that step. It does two things. First, it shows how to summarize what we observed. Second, it tells us how far to trust that summary. The second thing is the harder one. We have seen only a part, and we want to speak about the whole.

This section builds the vocabulary used in the rest of the course. It has three parts.

- §1.1 splits statistics into two kinds. One kind describes the data we have. The other kind draws conclusions about data we do not have.
- §1.2 defines four basic terms: population, sample, parameter, statistic. It also defines how we draw a sample.
- §1.3 sorts data into types. It then shows how a data set can mislead us, even when the data set is very large.

None of this is hard. But every symbol in Sections 2–7 goes back to one of these ideas.

1.1 What Is Statistics?

Raw *data* are recorded observations. Examples are 250 daily closing prices, the answers of 800 survey respondents, and the defect counts of 40 batches. Data settle nothing on their own. They become *information* when we organize them so that a pattern shows. That information supports *decision-making* only after we judge how reliable the pattern is. Statistics does the last two steps. It works in two modes. Section 2 covers the first mode, and Sections 3–7 build the second.

Definition 1.1 (Descriptive and inferential statistics).

- (i) Descriptive statistics (敘述統計): *graphical and numerical procedures that organize and summarize a set of data.*
- (ii) Inferential statistics (推論統計): *procedures that use data from a sample to draw conclusions about the larger population it came from. The conclusions may be estimates, tests or forecasts.*

The two modes use much the same arithmetic. They differ in how far they reach.

Suppose you report the average return of the 30 stocks you hold. That is a _____ statement. It is a fact about those 30 stocks, and it is exactly right. Now suppose you draw 30 stocks at random from a market of 900. You use their average return to speak about the whole

market. That is an _____ statement. It is a claim about 900 stocks based on 30, so it can be wrong.

Inference always carries uncertainty. This course exists to measure that uncertainty, not to hide it.

To measure it we need a probability (機率) model of how the sample was generated. We build that model in Sections 3–6. To describe the model we first need names for the objects it connects. §1.2 defines them, and we come back to the whole picture there.

1.2 Population and Sample

Definition 1.2 (Population and sample).

- (i) The population (母體) is the _____ set of items of interest, of size N (possibly infinite).
- (ii) A sample (樣本) is an _____ of the population, of size n .

The population is fixed by the *question*, not by the data you happen to have. Suppose you want to know how Taiwanese households invest. Then the population is all Taiwanese households. One brokerage's clients are not that population. Interviewing more of them does not fix this. People who open a brokerage account are already a special group, and households with no account never reach the list at all.

Now suppose we want a number that describes the whole population, such as the average holding of all Taiwanese households. Only one method gives that number for certain. We measure every member of the population. This is a *census* (普查).

A census is rarely available. Three obstacles recur.

- (1) **Cost and time.** Taiwan's population and housing census is run by the government once every ten years. No firm can repeat that to answer a question this month.
- (2) **Measuring destroys the item.** To learn how long a light bulb lasts, we must run it until it fails. If we test every bulb, we have nothing left to sell.
- (3) **The population cannot be listed at all.** This case matters most, and it has its own name.

A population does not have to be a set of things that already exist. Take the daily return on a stock. We may hold 250 trading days of data. But those 250 days are not what we care about. We care about the mechanism that produced them. The population is every return that mechanism could produce. It has no list. It is infinite, and most of it has not happened yet. A population like this is called _____. No census can reach it, because most of it does not exist yet.

So we sample. We measure a part and reason about the whole. Sampling is not a second-best choice. For most questions it is the only method there is. The rest of this course is about using it honestly.

We now have two kinds of number to keep apart. One describes the population and is the one we want. The other is computed from the sample and is the one we can actually get. Confusing them is the most common beginner's error.

Definition 1.3 (Parameter and statistic).

- (i) A parameter (母數 / 參數) is a numerical measure of a population characteristic. It is _____, and usually unknown.
- (ii) A statistic (統計量) is a numerical measure computed from a sample. We know its value once we have the sample, but it _____.

This asymmetry matters, so read it carefully. The parameter is the mean return μ of all 900 stocks. It is a fixed number, and we do not know it. The statistic is the mean $\bar{x}$ of your 30 stocks. We know it exactly. But it is not fixed: another 30 stocks give another $\bar{x}$. So the quantity we *want* is constant and hidden. The quantity we *have* is visible and random. Section 7 treats a statistic as a random variable and studies its distribution.

We now have all four objects, so we can draw the whole picture. Figure 1 is the logic of the course. We fix a population. We draw a sample by a mechanism we control. We compute a statistic. Then we reason back to the parameter. The forward step is chance. The backward step is inference. Probability measures the gap between them.

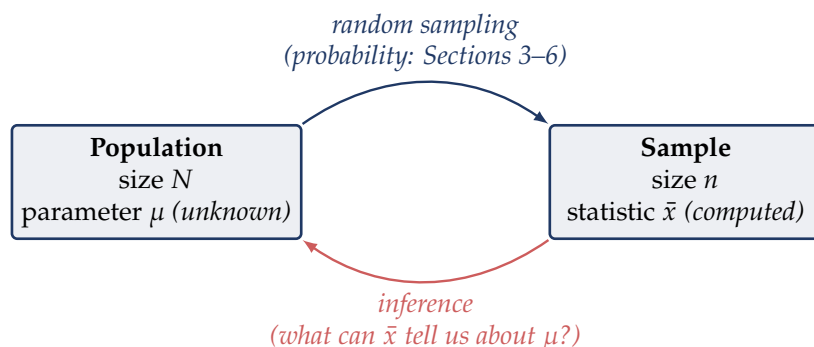


Figure 1: The inference loop. We can model the forward arrow, because we know how the sample was drawn. Probability then tells us how $\bar{x}$ behaves. The backward arrow is the one we want. It is only as good as our model of the forward arrow. A sample statistic is usually not equal to the parameter, and we can never check whether this sample happened to land on it. So the honest question is not “is it right?” It is “how far off can it plausibly be?”

The forward arrow is trustworthy only when we know how the sample was drawn.

Definition 1.4 (Simple random sample).

A simple random sample (簡單隨機樣本) of size n is one selected so that

- (i) every member of the population is _____ to be chosen;
- (ii) no member's selection influences another's; and
- (iii) every possible sample of size n is equally likely.

It is usually just called a random sample (隨機樣本).

We add one note here, because we need it later. In practice we draw *without replacement* (取後不放回) from a finite population. Then the draws are not strictly independent. Once we take a member it cannot appear again, so condition (ii) holds only approximately. When n is small

relative to N , say n/N below 5%, the dependence is negligible. We then treat the observations as independent. Every formula in Sections 3–7 is derived under that assumption.

The definition constrains the _____, not the result. A random sample can contain only large firms by bad luck. That makes it unlucky, not bad. We know the mechanism, so we can compute how often such bad luck strikes. This is the trade-off. Randomness does not guarantee a representative sample. It guarantees a *quantifiable* chance of an unrepresentative one. A convenience sample (便利樣本) is different. It takes whoever answers the phone. We do not know how that mechanism behaves, so there is nothing to compute.

A common practical alternative is *systematic sampling* (系統抽樣). Order the list. Pick a random start among the first j items. Then take every j th item. The sampling interval (抽樣間隔) is

$$j = \quad .$$

This is cheap, and it usually behaves like a random sample. But that holds only if the ordering is unrelated to the variable we study. The danger is a _____ pattern in the list with period j , or with a period that divides j . Every selected item then sits at the same point of the cycle, and the sample is systematically unlike the population.

Example 1.5. A brokerage's client list holds $N = 8,000$ accounts, ordered by the date each was opened. An analyst wants a systematic sample of $n = 200$.

- Find the sampling interval j .
- With random start 7, give the first three accounts selected and the last one.
- Describe one ordering of the list that would make the sample misleading.

Even a flawless procedure usually gives a number that differs from the parameter. That gap has two sources. They deserve separate names, because only one of them is a statistical problem.

Sampling error (抽樣誤差) comes from observing n items rather than all N . It is unavoidable, and it is not a mistake. It is also _____. We know the mechanism, so we can say how large the error is likely to be. It also shrinks as n grows. Sections 3–7 study it in detail.

Nonsampling error (非抽樣誤差) is everything that would remain even in a complete _____. Three sources recur:

- (1) **Wrong frame.** The sampling frame (抽樣底冊) you draw from leaves out part of the target population. Landline numbers, your own customers, and the funds that still exist are all examples.
- (2) **Inaccurate response.** The question is ambiguous, or it pushes the respondent toward one answer, or it is too sensitive for an honest answer. “How much have you stolen from this plant?” will not give reliable data.
- (3) _____. Those who decline are rarely a random subset. You end up with a random sample of the population *willing to respond*.

The warning follows at once. Sampling error shrinks toward zero as the sample grows. The *bias* (偏誤) from a nonsampling error does not. A larger sample shrinks the sampling error around whatever the flawed procedure measures. But the procedure still measures the wrong thing. A biased survey of a million people is no better than one of a thousand.

1.3 Data Categories and Data Issues

Before you choose any procedure, you must know what kind of data you have. The kind of data decides which summaries are meaningful at all.

Definition 1.6 (Variable; categorical and numerical).

A variable is a characteristic that may differ across the population. It is one of two types:

- (i) **categorical** (類別變數): *the values are groups or labels, such as industry, gender, or a 1–5 satisfaction rating. This holds even when we store them as numeric codes.*
- (ii) **numerical** (數值變數): *the values are quantities, obtained by counting or by measuring. A numerical variable is discrete (離散) if we can list its values one by one. It is continuous (連續) if it can take any value in an interval.*

So the number of trades a client places in a month is _____. The market value of their portfolio is _____. There are two common traps here.

- **Codes are not quantities.** A satisfaction rating stored as 1, . . . , 5 has numeric entries. But “5” is only a label for “very satisfied”. We chose that label by convention, and “A” would work just as well. A coded category is still a _____ variable.
- **Rounding is not discreteness.** We record a share price to two decimals, so the records form a discrete set. But the price itself varies over a continuum, so the variable is continuous. The classification refers to the quantity, not to the precision of the record.

A second classification asks how much arithmetic the numbers support. It is due to Stevens (1946), and it recognizes four measurement levels (衡量尺度).

Definition 1.7 (Measurement levels).

- (i) Nominal (名目尺度): *labels with no order.*
- (ii) Ordinal (順序尺度): *we can _____ the values, but the differences carry no fixed meaning.*
- (iii) Interval (等距尺度): *the differences are meaningful, but the zero point is _____.*
- (iv) Ratio (比率尺度): *the differences are meaningful and the zero is _____. So ratios of values make sense.*

Nominal and ordinal data are qualitative (質性資料/定性資料); interval and ratio data are quantitative (量化資料/定量資料).

The two classifications fit together. Nominal and ordinal are the levels of *categorical* variables. Interval and ratio are the levels of _____ ones. So classifying a variable takes two steps. First decide categorical or numerical. Then decide which of its two levels. Most mistakes happen at the last two levels. The test is simple: ask whether the zero is a _____ or a genuine absence. Someone chose the zero on the Celsius scale, so 20°C is not twice as warm as 10°C. Zero dollars is a real absence, so NT\$2m really is twice NT\$1m.

Level	Supports	Does not support	Example
Nominal	counting, mode	ranking, differences	industry sector
Ordinal	ranking, median	differences, means	bond rating (AAA, AA, ...)
Interval	differences, mean	ratios	temperature (°C), calendar year
Ratio	differences, ratios	—	portfolio value, trading volume

Table 1: The four measurement levels and the arithmetic each one allows. The levels are cumulative as you read *downward*. Each row supports everything the row above supports, and one thing more. Read the table for what it forbids: it tells you which computations are *meaningless*.

This matters most with ordinal data coded as numbers. A satisfaction scale runs from 1 (very dissatisfied) to 5 (very satisfied). You can compute the average of those numerals. But the gap between “4” and “5” need not equal the gap between “1” and “2”. The codes carry order, not _____. So the mean of such codes has no calibrated meaning. The _____ and the frequency distribution are the defensible summaries. The mean assumes an equal spacing that the scale never established.

Example 1.8. A bank stores three fields for each of its 500 corporate clients:

- (i) the client's credit rating (AAA, AA, ..., C);
- (ii) the calendar year in which the account was opened;
- (iii) the balance on the account, in NT\$.

For each computation below, say whether it is meaningful, and why.

- (a) The mean credit rating.
- (b) The difference between two clients' opening years.
- (c) The ratio of two clients' balances.
- (d) The ratio of two clients' opening years.

We close with a failure that no technique repairs. Both problems below are nonsampling errors. They turn on *which* items entered the data set, not on how many.

Selection bias (選擇偏誤) occurs when the sample admits items for a reason connected to the variable we study. The sampled population then differs systematically from the target. Suppose a brokerage wants the average annual return of *all* retail investors in the country. It estimates that return from the accounts enrolled in its premium research service. Subscribers trade more and hold larger accounts than the typical investor, so their returns differ systematically. The _____ the brokerage sampled is its subscribers, and that is not the target population. The sample can be huge, and the bias survives untouched.

The everyday version is _____. Online reviews are written by customers who chose to write one. So the average rating describes opinionated customers, not customers. The same mechanism appears in finance wherever reporting is voluntary. A hedge fund index built from funds that choose to report describes the funds *willing to report*. It does not describe "hedge funds".

Selection bias also works in the time dimension. This version matters most for the return data you meet in finance. A fund database holds the funds that *still exist*. Funds that did badly

were closed or merged away, and their returns went with them. An average over the survivors measures a group that was selected, after the fact, for having done well. The data show no sign of this. We call it _____ bias.

Example 1.9. Ten years ago a database tracked 100 equity funds. Today 70 still operate. The other 30 were closed or merged away during the decade. Over the ten years the 70 survivors averaged 8.0% per year. The 30 that disappeared averaged 2.0% per year while they existed. A magazine uses today's database and reports: "the average equity fund returned 8.0% a year over the decade."

- (a) Average over all 100 funds the database began with. Weight each fund equally, and let each contribute its own average annual return. What is the cohort's mean annual return?
- (b) By how much does the magazine's figure overstate it?
- (c) Which population does 8.0% really describe, and why is it useless to an investor choosing a fund today?

Exercises

Exercise 1.1. A bank surveys a random sample of its credit-card holders. For each variable, state whether it is categorical or numerical; if categorical give the measurement level, if numerical say discrete or continuous.

- (a) The type of card held (Visa, Mastercard, JCB).
- (b) Satisfaction with the mobile app (1: very dissatisfied, ..., 5: very satisfied).
- (c) The number of late payments made last year.
- (d) The amount charged to the card last month, in NT\$.

Exercise 1.2. A researcher studies the 1,200 firms listed on the Taiwan Stock Exchange. She draws 60 at random and finds their mean return on equity is 9.4%; the mean ROE of all 1,200 firms is unknown.

- (a) State the population and its size N .
- (b) State the sample and its size n .
- (c) Is 9.4% a parameter or a statistic?
- (d) Suppose she drew a different 60 firms. Which number would change: the mean ROE of the 1,200, or the mean of her sample? What does this say about which one is random?

Exercise 1.3. An auditor must examine $n = 120$ invoices from a file of $N = 3,600$, arranged in the order they were issued, and samples systematically.

- (a) Find the sampling interval j .
- (b) With random start 17, give the first three invoices selected and the last one.
- (c) The firm issues invoices in a fixed daily cycle: one per branch, for its 30 branches, always in the same order. What goes wrong, and how would you fix it?

Exercise 1.4. Classify each as sampling error or nonsampling error, with a one-sentence reason.

- (a) The mean income of the 500 sampled households differs from the mean income of all households in the city.
- (b) Forty percent of the households contacted refuse the income question.
- (c) The survey frame is a list of landline telephone numbers.
- (d) Respondents systematically understate their income.

In which of these is the bias reduced by contacting 5,000 households instead of 500?

Exercise 1.5. A data vendor's database began five years ago with 250 hedge funds. Today 200 remain, averaging 7.5% per year over the five years; the 50 that closed averaged 1.5% per year while they operated.

- (a) What mean annual return does the vendor report using only the funds now in the database?

- (b) *Average over all 250 original funds. Weight each fund equally, and let each contribute its own average annual return. What is the cohort's mean annual return?*
- (c) *State the size and direction of the bias.*
- (d) *Would tracking 1,000 surviving funds instead of 200 give a better figure? Explain.*

Exercise 1.6. *An online retailer shows a mean customer rating of 4.6/5, computed from the reviews it received. Separately, a hedge fund index is computed each month from the returns of funds that voluntarily report to the provider.*

- (a) *Name the bias in each case, and state precisely which population each number describes.*
- (b) *The 4.6 is an average of an ordinal variable. Explain why that is a second, separate problem.*
- (c) *Propose one concrete change to the retailer's data collection that removes the bias in (a), and say honestly what it costs and what remains.*